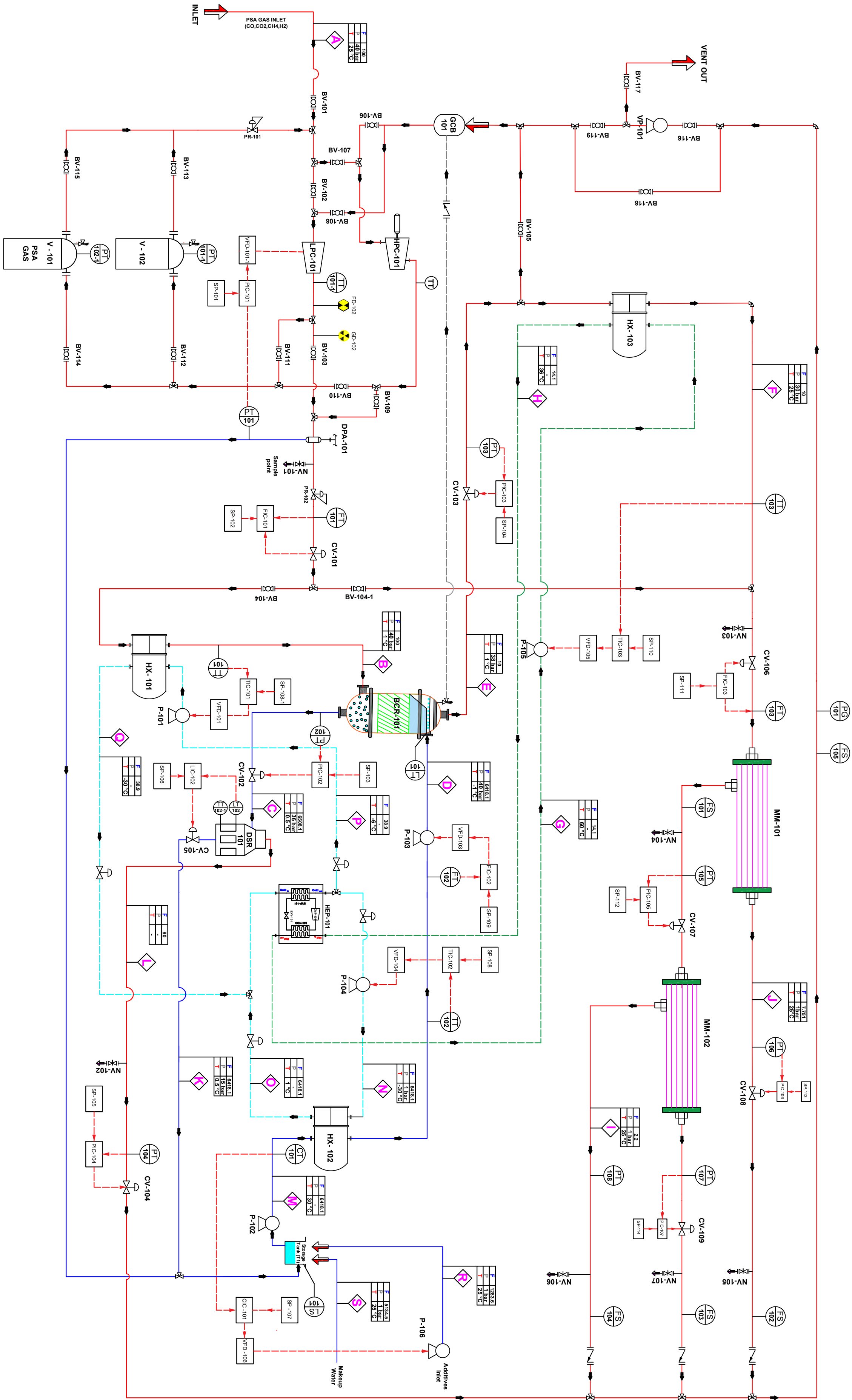


PIPING AND INSTRUMENTATION DIAGRAM OF PILOT PLANT FOR H₂ AND CH₄ FROM PSA OFF GAS REACTOR ALONE PRESSURIZED HIGH DENSITY HYDRATE



PFD Legends

- Gas Stream
- Water Loop
- Hot Transfer Fluid
- Cold Transfer Fluid
- Pressure Relief Discharge Line

LIC - Level Indicating Controller

- SP - Set Point
- LPC 101 - Liquid Piston Compressor
- DPA - Pulsation Damper
- BCR 101 - Bubble Column Reactor
- DSR 101 - Dissociator
- HEP 101 - Heat Pump
- HX 101 & 102 - Heat Exchanger
- MM 101 & 102 - Membrane Module

FIC - Flow Indicating Controller

- PIC - Pressure Indicating Controller
- P 101 to 105 - Pumps
- VP 101 - Vacuum Pump
- V 101 to 104 - Pressure Vessels
- GCB 101 - Gas Collection Bag
- PRV - Pressure Regulator Valve
- CV - Control Valve
- BV - Ball Valve

VFD - Variable Frequency Drive

- TIC - Temperature Indicating Controller
- PT - Pressure Transmitter
- TT - Temperature Transmitter
- FT - Flow Transmitter
- FS - Flow Sensor
- LS - Level Sensor
- CS - Concentration Sensor

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